

AI-CROP SHIELD

AI-Based Early Detection, Risk Prediction and Targeted Management of Crop Diseases and Pests

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Project Name	AI-CROP SHIELD
Team Name	NEBULA
Institution	K. S. Institute of Technology, Bengaluru
Event	Smart India Hackathon 2026
Project Type	AI + IoT + GIS + Mobile/Web Decision-Support Platform

1. Executive Summary

AI-Crop Shield is a proposed intelligent agriculture platform designed to help farmers detect crop diseases and pest infestations at an early stage, predict outbreak risk, identify geographical hotspots, and receive practical management recommendations in a local language.

The system combines computer vision, machine learning, IoT sensing, GIS mapping and a farmer-facing mobile/web interface. A farmer can capture a crop image using a smartphone, field camera or drone. The image is processed by a computer-vision model to identify probable disease or pest symptoms and estimate severity. Environmental information such as temperature, humidity, soil moisture and rainfall is collected through an IoT layer. These data sources are combined to estimate outbreak risk before the problem spreads widely.

The platform then maps affected or high-risk locations using GIS and provides localized alerts and crop-specific Integrated Pest Management (IPM) recommendations. The objective is to move from reactive treatment after visible damage to preventive and targeted crop protection, while reducing unnecessary pesticide use and improving access to timely agricultural guidance.

2. Problem Statement

Farmers may face delayed identification of crop diseases and pest infestations, rapidly changing weather conditions, limited access to expert advice, difficulty identifying affected areas, and unnecessary or untargeted pesticide application.

In many practical situations, disease symptoms become noticeable only after crop damage has progressed. Weather and field conditions can also influence disease and pest development. A farmer may therefore need a system that does more than classify a crop image: it should connect visual symptoms with environmental conditions, communicate risk, identify affected locations and provide an understandable action plan.

AI-Crop Shield addresses this need through an integrated workflow:

Detect → Predict → Map → Alert → Act.

3. Objectives

1. Detect probable crop diseases and pest infestations from crop images.
2. Estimate the severity of the detected condition.
3. Collect field-level environmental parameters using IoT sensors.
4. Combine image and environmental data for early outbreak-risk prediction.
5. Create GIS-based hotspot maps for affected and high-risk areas.
6. Provide alerts and recommendations in regional/local languages.
7. Support crop-specific Integrated Pest Management (IPM).
8. Encourage treatment of affected areas instead of unnecessary whole-field pesticide application.
9. Maintain historical field data for future analysis and model improvement.
10. Design the prototype so it can be extended from one validated crop to multiple crops and regions.

4. Proposed Solution

AI-Crop Shield consists of four connected layers:

A. Data Acquisition Layer

- Crop images from a smartphone, field camera or drone.
- IoT measurements including temperature, humidity, soil moisture and rainfall.
- Optional location information for field-level GIS mapping.

B. AI/ML Layer

- Image preprocessing using OpenCV.
- Deep-learning-based disease/pest classification using TensorFlow or PyTorch.
- Severity estimation.
- Risk prediction using image features and environmental observations.

C. Cloud, Database and GIS Layer

- Backend APIs using Flask or FastAPI.

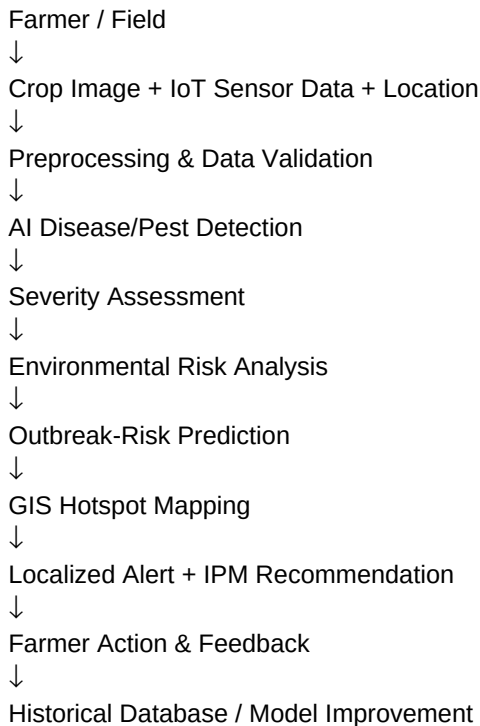
- Cloud/database storage using Firebase and/or PostgreSQL.
- GIS visualization using Google Maps API, Leaflet or another map service.
- Historical records for fields, observations, predictions and alerts.

D. User Interaction Layer

- Farmer mobile/web interface using React/React Native or Flutter.
- Local-language alerts and recommendations.
- Risk maps and field history.
- Dashboard for monitoring affected areas and reports.

5. System Architecture

The proposed data flow is:



The architecture is modular so that image classification, risk prediction, mapping and user-interface components can be improved independently.

6. Technical Approach

Step 1 — Data Acquisition

The farmer captures a crop image through a mobile application, camera or drone. IoT sensors can simultaneously provide temperature, humidity, soil moisture and rainfall observations.

Step 2 — Image Processing

The image is validated, resized and enhanced where necessary. OpenCV can be used for preprocessing such as resizing, normalization and noise reduction.

Step 3 — Disease/Pest Detection

A trained computer-vision model identifies the probable disease or pest. The model can be based on a convolutional neural network or another suitable vision architecture. The system also reports a confidence value and severity category.

Step 4 — Environmental Analysis

Environmental measurements are evaluated along with image-based evidence. The risk model can learn relationships between crop condition, weather/soil parameters and observed disease or pest occurrence.

Step 5 — Risk Prediction

The system generates an outbreak-risk level such as Low, Medium or High. The prediction is intended as a

decision-support signal, not as a substitute for an agricultural expert.

Step 6 — GIS Hotspot Mapping

Location-tagged observations are visualized on a map. High-risk clusters can be highlighted so that farmers and authorized stakeholders can prioritize inspection and intervention.

Step 7 — Localized Alert and Recommendation

The application provides an understandable alert and crop-specific IPM guidance in the selected local language. Recommendations should prioritize appropriate, targeted and safe intervention.

Step 8 — Feedback and Historical Data

User observations, confirmed diagnoses and intervention outcomes can be stored for future analysis and model refinement.

7. Technology Stack

Software:

- Python — AI/ML development and backend integration
- TensorFlow / PyTorch — deep-learning model development
- OpenCV — image processing
- Flask / FastAPI — backend APIs
- React / React Native or Flutter — farmer-facing application
- Firebase / PostgreSQL — data and user storage
- Google Maps API / Leaflet — GIS and hotspot visualization
- Matplotlib — charts and analytical visualization

Hardware:

- Smartphone / field camera / drone — image acquisition
- ESP32 / Arduino — sensor data collection and communication
- Temperature and humidity sensor such as DHT22/BME280 — environmental monitoring
- Soil-moisture sensor — soil-condition monitoring
- Rainfall sensor or external rainfall data source — rainfall information
- Cloud server — AI processing, APIs and data storage

The exact components can be changed during prototyping depending on cost, availability and field-testing requirements.

8. Implementation Plan

Phase 1 — Prototype Foundation

- Select one crop and a limited set of disease/pest classes.
- Prepare or obtain a properly labeled image dataset.
- Build the initial image-classification/detection pipeline.
- Create a simple farmer interface.

Phase 2 — IoT Integration

- Connect ESP32/Arduino with environmental sensors.
- Send sensor readings to the backend/cloud.
- Store timestamped field observations.

Phase 3 — Risk Prediction

- Combine image predictions and environmental data.
- Build a baseline risk model.
- Define transparent Low/Medium/High risk thresholds for the prototype.

Phase 4 — GIS and Alerts

- Store location-tagged observations.
- Display field/village-level hotspots.
- Implement localized notifications and IPM guidance.

Phase 5 — Validation

- Test the complete workflow under controlled and, where feasible, field conditions.

- Measure model performance using appropriate validation data.
- Measure latency, sensor reliability, usability and alert accuracy.
- Document limitations and improve the prototype iteratively.

9. AI/ML Methodology

The AI pipeline can be implemented as follows:

Input image

- quality check
- preprocessing
- feature extraction
- disease/pest classification
- severity estimation
- confidence calculation.

For a first prototype, a transfer-learning approach can be considered because it can reduce training requirements compared with building a deep model entirely from scratch. Candidate architectures may include MobileNet, EfficientNet or other suitable CNN-based models, selected after experimentation.

For risk prediction, image-derived signals can be combined with environmental features such as temperature, humidity, soil moisture and rainfall. A baseline model may use logistic regression, random forest, gradient boosting or a small neural network, followed by comparison using validation data.

Important evaluation metrics include accuracy, precision, recall, F1-score and confusion matrix for classification. For risk prediction, precision/recall, ROC-AUC where appropriate, and calibration/error analysis can be considered. These metrics should be reported only after actual experiments are completed.

10. GIS Hotspot Mapping

Each observation can be associated with a field or location identifier. The GIS module can display:

- Low-risk locations
- Medium-risk locations
- High-risk locations
- Recently reported cases
- Disease/pest type
- Severity
- Time of observation

Spatial clustering can help identify areas requiring inspection. The map is intended to support prioritization rather than replace field verification. Location privacy and access controls should be considered when storing farmer or field information.

11. Local-Language Recommendation Module

The system is designed to communicate the result in a language that is understandable to the farmer. A recommendation can contain:

- Detected condition
- Severity/risk level
- Immediate action
- Preventive action
- IPM option
- When to seek expert confirmation

For example, the application may translate a structured recommendation into the selected regional language rather than generating unrestricted medical-style or agricultural claims. The recommendation database should be reviewed by appropriate agricultural experts before real-world deployment.

12. Innovation and Uniqueness

The central innovation is the integration of multiple signals and actions in a single workflow rather than relying only on image classification.

1. Multimodal intelligence — combines crop images with environmental information.
2. Early risk prediction — aims to identify risk before widespread visible damage.
3. GIS hotspot intelligence — converts individual observations into location-level risk information.
4. Localized communication — makes alerts more accessible through regional languages.
5. Targeted intervention — supports crop-specific IPM and affected-area treatment.
6. Historical learning — stores field observations for future prediction and analysis.
7. Scalable architecture — starts with one crop for controlled validation and can later expand to multiple crops and regions.

13. Expected Outcomes

The prototype is expected to demonstrate:

- Automated crop-image disease/pest identification.
- Severity categorization.
- Real-time or near-real-time environmental monitoring.
- Combined image + environmental risk assessment.
- GIS visualization of affected/high-risk locations.
- Local-language farmer alerts.
- Structured IPM recommendations.
- Historical field records and dashboard visualization.

These are intended prototype outcomes; quantitative improvement claims should be made only after field or benchmark evaluation.

14. Advantages

- Earlier access to crop-health information.
- Integrates image and field-environment signals.
- Provides location-aware risk visualization.
- Improves accessibility through local-language communication.
- Supports targeted rather than indiscriminate intervention.
- Creates a digital history of crop-health observations.
- Can be scaled to additional crops and regions after validation.

15. Limitations and Risk Management

1. Image quality can affect model predictions.

Mitigation: image-quality checks, augmentation and clear capture guidance.

2. Similar symptoms can lead to classification errors.

Mitigation: confidence thresholds and expert verification for uncertain cases.

3. Sensor failure or missing readings can affect risk prediction.

Mitigation: sensor-health checks, timestamps and missing-data handling.

4. A model trained on one region/crop may not generalize to another.

Mitigation: region/crop-specific validation and periodic retraining.

5. GIS data may contain sensitive field information.

Mitigation: access control, minimum necessary data collection and secure storage.

6. Recommendations can be misapplied.

Mitigation: expert-reviewed IPM content, clear warnings and escalation to agricultural experts where required.

16. Future Scope

- Support multiple crops and disease/pest classes.
- Add multilingual voice interaction.
- Integrate satellite or drone imagery for larger-area monitoring.

- Add weather forecasting data for forward-looking risk models.
- Introduce federated or privacy-preserving learning where appropriate.
- Add expert verification and feedback loops.
- Develop district/state-level dashboards for authorized agricultural stakeholders.
- Integrate farmer advisories and government agricultural information where permitted.
- Conduct larger field trials and quantify reduction in crop loss or unnecessary pesticide use.

17. Prototype Demonstration Flow

During the SIH prototype demonstration, the team can show one complete scenario:

1. Open the farmer application.
2. Capture/upload an image of an affected leaf.
3. Display the AI prediction, confidence and severity.
4. Show live/simulated IoT readings.
5. Generate the combined risk level.
6. Open the GIS map and show the affected/high-risk hotspot.
7. Display a local-language alert.
8. Show the crop-specific IPM recommendation.
9. Open the dashboard and show that the observation has been stored.

This sequence demonstrates the complete value chain instead of presenting disconnected technologies.

18. GitHub Repository and YouTube Prototype

For the final SIH submission, the team should provide two working links:

GitHub Repository:

[INSERT ACTUAL GITHUB REPOSITORY LINK]

Recommended repository contents:

- Source code
- README with setup instructions
- Dataset description and licensing information
- Model/training documentation
- Hardware/IoT code
- Backend/API code
- Frontend/mobile code
- Architecture diagram
- Screenshots
- Test results
- Team contribution information

YouTube Prototype:

[INSERT ACTUAL YOUTUBE DEMO LINK]

Recommended video structure:

- 0:00–0:20 Problem
- 0:20–0:45 Farmer image capture
- 0:45–1:10 AI detection
- 1:10–1:35 IoT/environmental data
- 1:35–2:00 Risk prediction
- 2:00–2:25 GIS hotspot map
- 2:25–2:50 Local-language alert and IPM
- 2:50–3:00 End result.

19. Conclusion

AI-Crop Shield proposes an integrated decision-support platform for early crop-health monitoring. By connecting computer vision, IoT sensing, risk prediction, GIS mapping and localized recommendations, the system is designed to help convert scattered field observations into actionable information.

The prototype should first be validated on a defined crop and a limited set of conditions. Once accuracy, reliability and usability are demonstrated, the architecture can be extended to additional crops, regions and data sources. The overall approach focuses on preventive and targeted crop protection rather than waiting for widespread damage before action is taken.

APPENDIX — MATERIALS TO ADD BEFORE FINAL SIH SUBMISSION

- Actual GitHub repository URL
- Actual YouTube prototype URL
- Final problem statement ID and exact official wording
- Team member names, roles and contact details as required by the SIH format
- Prototype screenshots
- Hardware photographs
- Real farmer/problem photographs with source/permission details where applicable
- Actual model evaluation results and dataset details
- System architecture diagram
- Testing/validation results